Motegi's (D) without Motegi's premises: a short proof via the left Demazure recursion

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Corresponds to WIP commit `7cdb983` (`grandpa-rick/work-in-progress`, `notes/`).
Follows: `notes/2026-10-22-addendum-theorem-D-scope.pdf` (WIP `4c175b4`), which promised this proof.
Sources: `proofs/2026-10-22-dfree-biorthogonality.md`; registry `proofs/registry/q488-converse.json`; checks: WIP `scripts/prove250/` (`chk1.py`, `nclosed.py`, `nform.py`, `ctrl.py`).

Summary

- **What (D) means is classical.** (D) is the fixed-point (Atiyah–Bott localisation) form of the K_T duality on $LG(n,2n)$ between Schubert structure sheaves $[\mathcal{O}_{X_\kappa}]$ and opposite boundary-ideal sheaves $[\mathcal{O}_{X^\lambda}(-\partial X^\lambda)]$. That duality is classical (cf. Brion; Graham–Kumar; I haven't pinned the IDs). This note does *not* claim a new biorthogonality.
- **The explicit formula is Motegi's.** The explicit fixed-point coefficients $\rho_{\kappa L}$ of (D) are Motegi's (arXiv 2610.15344).
- **What is ours.** (i) A short, elementary proof of (D) (Theorem D$'$) that uses neither of the two premises of Motegi's argument I could not audit: (P1) the type-C branching/factorisation, and (P2) the $\hat{\Phi}$-inverse/RTV input. It also uses no Ikeda–Naruse, no K0, and no Atiyah–Bott. (ii) Corollary K, a closed form for the C_n Demazure localisation coefficients: $a_L^\kappa = \Psi_{\kappa^c,0}(L^c)/\prod_{k\le l}(1-(u_k u_l)^{-1})$.
- **Owner caveat.** Lemmas $\mathrm{A}^0/\mathrm{B}^0$ below very likely are the left Demazure–Lusztig operators of Mihalcea–Naruse–Su (arXiv 2008.12670), evaluated at fixed points. I haven't read that paper yet. The proof here is self-contained, but I am not claiming the operators as new.
- **Grades.** Theorem D$'$, Lemma N and Lemmas $\mathrm{A}^0/\mathrm{B}^0$ are proved. K0 (Ψ^0 = Ikeda–Naruse GQ) stays computed. The 12 nodes I downgraded after your UID 435 are restored to proved (§5).

1 Setup

This uses the notation of the addendum and of `prove249c/eng.py`. Let $w_1,\dots,w_n$ be variables. For $\nu=(\nu_1<\cdots<\nu_r)\subseteq[n]$ and $M\subseteq[n]$,

$$\Psi_{\nu,\theta}(M)=\sum_{\substack{h:[r]\hookrightarrow M\\ h_j\ge\nu_j}} T^0_\nu(h)\prod_j x_j^{\theta_j},\qquad x_j=w_{h_j},$$

$$T^0_\nu(h)=\prod_j(1-x_j^{-2})\,\pi_{\nu_j}(x_j)\prod_{j<k}\frac{1-1/(x_jx_k)}{1-x_j/x_k}\prod_{t\in M\setminus h([r])}\prod_j\frac{1-1/(w_tx_j)}{1-w_t/x_j},\qquad \pi_k(y):=\prod_{k'<k}(1-w_{k'}/y).$$

(The proof file writes $\rho_k(y)$ for $\pi_k(y)$. I renamed it here so it does not clash with $\rho_{\kappa L}$.)

Further notation:

- $\gamma_\nu:=\Psi_{\nu,0}$ is the untwisted function.
- $F_\kappa(M):=w_M^{2r}\Psi_{\kappa,(\kappa_j-2j)}(M)$ is the explicit basis, and $g_\kappa:=w^2F_\kappa$.
- $\mathrm{cc}(I):=\prod_{i\in I}(-1)^i\prod_{k\le i}w_k$.
- $A_\kappa:=(-1)^{\Sigma\kappa}\prod_{k\in\kappa}\prod_{j\le k}w_j^{-1}$. Note that $\mathrm{cc}(\kappa)A_\kappa=1$.

Motegi's (D) (q$\to\infty$ face, as in `2026-10-18-q488-d-infinity.md`, Prop. 2):

$$\sum_{L\subseteq[n]}\rho_{\kappa L}\,F_\lambda(L)=\delta_{\kappa\lambda},\qquad \rho_{\kappa L}:=\frac{\mathrm{cc}(L)}{\mathrm{cc}(\kappa)}\cdot\frac{\gamma_{\kappa^c}(L^c)}{F_L(L)\,\gamma_{L^c}(L^c)}.$$

W-action (as in §1bis of the window file):

- s_m $(m<n)$ swaps $w_m\leftrightarrow w_{m+1}$ and swaps the labels $m\leftrightarrow m+1$. Its root factor is $e^{-\alpha}=w_m/w_{m+1}$.
- σ sends $w_1\mapsto 1/w_1$ and $L\mapsto L\triangle\{1\}$. Its root factor is $e^{-\alpha}=w_1^{-2}$.
- On functions of subsets, $(s\cdot f)(M):=s\big(f(sM)\big)$. Both actions commute with complementation.

Demazure localisation coefficients. Set $a^{\emptyset}_L=\delta_{L\emptyset}$. For $\kappa=s\kappa'$ with $\Sigma\kappa=\Sigma\kappa'+1$,

$$a^\kappa_L=\frac{a^{\kappa'}_L-e^{-\alpha}\,s(a^{\kappa'}_{sL})}{1-e^{-\alpha}}.\tag{1}$$

The admissible steps are s_m with $m\in\kappa'$, $m+1\notin\kappa'$, and σ with $1\notin\kappa'$. Every $\kappa\neq\emptyset$ is reached: if $1\in\kappa$, drop it; otherwise lower $\min\kappa$.

Input from §1bis (already Motegi-free; grade proved). The abstract Theorem D from the addendum: $\Lambda_\kappa:=A_\kappa\sum_L a^\kappa_L\,w_L^2\,(\cdot)|_L$ satisfies $\Lambda_\kappa(F_\lambda)=\delta_{\kappa\lambda}$. Its proof is the induction from Lemmas A_m/B on g, with base $\Lambda_\emptyset(g_\lambda)=F_\lambda(\emptyset)=\delta_{\lambda\emptyset}$. The circular closing line "$\tilde\Lambda_\kappa=\ell_\kappa$ by uniqueness" is *not* used.

Alternant form (Lemma 0). Group the injections by image H. The constraint $h_j\ge\nu_j$ can be dropped, because $\pi_{\nu_j}(w_{h_j})=0$ when $h_j<\nu_j$. Then

$$\gamma_\nu(M)=\sum_{H\subseteq M,\ |H|=r}S_0^M(H)\,A^0_\nu(x_H),\qquad A^0_\nu(x):=\frac{\det[x_i^{j-1}\pi_{\nu_j}(x_i)]}{V(x)},$$

where $V(x)=\prod_{i<k}(x_k-x_i)$ and

$$S_0^M(H)=\prod_{h\in H}(1-x_h^{-2})\prod_{h<h'}\Big(1-\frac{1}{x_hx_{h'}}\Big)\prod_{t\in M\setminus H,\,h\in H}\frac{x_h-1/w_t}{x_h-w_t}.$$

2 Lemma N: the one-point factor is the inverse Euler class

For $L \subseteq [n]$ put $u_k = w_k^{-1}$ if $k \in L$ and $u_k = w_k$ if $k \notin L$. This is the fixed point v_L read in w-coordinates.

Lemma 1 (N). *$w_L^2\, F_L(L)\, \gamma_{L^c}(L^c) = \mathrm{cc}(L) \prod_{1\le k\le l\le n}(1-(u_k u_l)^{-1})$. Hence*

$$N(L) := \frac{\mathrm{cc}(L)}{w_L^2 F_L(L)\gamma_{L^c}(L^c)} = \Phi(u), \qquad \Phi(u) = \prod_{k\le l}(1-(u_k u_l)^{-1})^{-1},$$

and $s\big(N(sL)\big) = N(L)$ for every simple reflection s.

Sketch. Both evaluations have exactly one injection, $h = \mathrm{id}$, so both are finite products. Sort the pairs $k \le l$ of $[n]$ into three kinds:

- both in $C = L^c$: these give the C-factors of $\gamma_C(C)$;
- both in L: $1 - w_k w_l$, rewritten from $(1 - w_l^{-2})$ and $(1 - 1/(w_l w_{l'}))$;
- mixed: from π, after cancelling denominators.

The leftover monomial has sign $(-1)^{\Sigma L}$, and w_{L_i} appears $r - i + 1$ times. That is exactly $\mathrm{cc}(L)$. Φ is symmetric in u:

- $u(s_m L, s_m w)$ is $u(L, w)$ with coordinates $m, m+1$ swapped;
- $u(L \triangle 1, \sigma w) = u(L, w)$.

Geometrically, $\Phi = 1/\mathrm{eul}_L$: the tangent weights of $LG(n, 2n)$ at v_L are $\{v_p\varepsilon_p + v_q\varepsilon_q\}_{p\le q}$. The proof doesn't use this. □

3 Lemmas A^0, B^0: left Demazure recursion for the untwisted γ_ν

These are the $\theta = 0$ mirror of §1bis Lemmas A_m/B (see the owner caveat in the Summary).

Lemma 2 (A_m^0). *Suppose $m < n$, $m+1 \in \nu'$, $m \notin \nu'$, and ν is ν' with $m+1$ replaced by m. Then $s_m \cdot \gamma_{\nu'} = e^\alpha \gamma_{\nu'} - (e^\alpha - 1)\gamma_\nu$, where $e^\alpha = w_{m+1}/w_m$.*

Proof. Substitute $H = s_m H_0$. Then S_0 and V are unchanged, and only the coefficients inside the π_k move.

- For $k \ne m+1$, π_k is unchanged: for $k \le m$ it doesn't involve w_m, w_{m+1}, and for $k \ge m+2$ it is symmetric in them.
- For $k = m+1$: $\pi'_{m+1} = \pi_m(1 - w_{m+1}/y) = e^\alpha \pi_{m+1} - (e^\alpha - 1)\pi_m$.

Exactly one column has $\nu'_j = m+1$, so multilinearity gives the claim. The new ν is still increasing because $m \notin \nu'$. □

Lemma 3 (B^0). *Let $z = w_1$. (a) If $1 \notin \nu$, then $\sigma \cdot \gamma_\nu = \gamma_\nu$. (b) If $1 \in \nu'$ and $\nu = \nu' \setminus 1$, then $\sigma \cdot \gamma_{\nu'} = z^2 \gamma_{\nu'} - (z^2 - 1)\gamma_\nu$.*

Proof. Put $\tau(y) = (y - 1/z)/(y - z)$, the tail-1 factor on a head of value y. Then $\sigma\tau = 1/\tau$, $\sigma\pi_k = \tau\pi_k$ for $k \geq 2$, and $\pi_1 = 1$.

(a). Take $M \not\ni 1$.

- Heads at 1 contribute 0, since every $\nu_j \geq 2$.
- $S_0^{M\cup 1}(H) = S_0^M(H) \prod_{h\in H} \tau(x_h)$.
- σ inverts those τ's and multiplies each row of A_ν^0 by τ. The two cancel.

(b). Reduction. Applying σ and using (a) for ν turns (b) at $M_0 \not\ni 1$ into (b) at $M_0 \cup 1$. So it suffices to show, for $M \not\ni 1$,

$$\sigma(\gamma_{\nu'}(M \cup 1)) = z^2\gamma_{\nu'}(M) + (1 - z^2)\gamma_\nu(M).$$

The columns of $A_{\nu'}^0$ are $f_1 = 1$ and $f_j = y^{j-1}\pi_{\nu'_j}$ for $j \geq 2$, and f_j/y is column $j - 1$ of A_ν^0.

(I) 1 a tail ($H \subseteq M$). After σ and dividing each row by τ, the term is $S_0^M(H) \det[1/\tau, f_2, \ldots, f_r]/V$. Since $1/\tau(y) - z^2 = (1 - z^2)\, y/(y - 1/z)$, the sum over H is $z^2\gamma_{\nu'}(M)$ plus $(1 - z^2) \sum_H S_0^M(H) \det[g, f_2, \ldots]/V$, where $g(y) = y/(y - 1/z)$.

Expand along column 1 with $H = H' \sqcup i$, and use $S_0^M(H' \cup i) = S_0^M(H')(1 - x_i^{-2}) \prod_{t\in T\setminus i} \frac{x_i - 1/w_t}{x_i - w_t} \prod_{k\in H'} \frac{x_k - x_i}{x_k}$, where $T = M \setminus H'$. The Vandermonde factors and $e_{H'} = \prod_{k\in H'} x_k$ cancel, and the remainder is

$$(1 - z^2) \sum_{H'} S_0^M(H') A_\nu^0(x_{H'}) \sum_{i\in T} R_i, \qquad R_i = \operatorname*{Res}_{u=w_i} \frac{h(u)}{u - 1/z}, \quad h(u) := \prod_{t\in T} \frac{u - 1/w_t}{u - w_t}.$$

(II) 1 a head ($H = \{1\} \sqcup H'$).

- After σ, head 1 has value $1/z$, and its row is $(1, 0, \ldots, 0)$.
- Head 1 contributes $(1 - z^2)$, the pair factors $\prod_k (1 - z/x_k)$, and the tail factors $h(1/z)$.
- Each row $k \in H'$ collects $\tau(x_k)(1 - z/x_k)/(x_k - 1/z) = 1/x_k$.

So the term is $(1 - z^2)h(1/z)S_0^M(H')A_\nu^0(x_{H'})$.

Residue identity. $h(u)/(u - 1/z) \sim 1/u$ at ∞, so its finite residues sum to 1. Hence $\sum_{i\in T} R_i = 1 - h(1/z)$. Adding (I) and (II), the $h(1/z)$ terms cancel and leave $(1 - z^2)\gamma_\nu(M)$. For $r = 1$, read empty determinants as 1 and set $\gamma_\emptyset \equiv 1$. □

4 Theorem D′ and Corollary K

Theorem 4 (D′)**.** $\sum_L \rho_{\kappa L} F_\lambda(L) = \delta_{\kappa\lambda}$ *for all* $\kappa, \lambda \subseteq [n]$. *Moreover* $\rho_{\kappa L} = A_\kappa\, a_L^\kappa\, w_L^2$.

Proof. Since $\mathrm{cc}(\kappa)A_\kappa = 1$ and by the definition of N,

$$\rho_{\kappa L} = A_\kappa\, w_L^2\, b_L^\kappa, \qquad b_L^\kappa := \gamma_{\kappa^c}(L^c)\, N(L).$$

We show $b^\kappa = a^\kappa$ by induction along the admissible chain that defines a^κ.

Base. $\gamma_{[n]}(L^c)$ needs n distinct heads, so it is 0 unless $L = \emptyset$. And $b_\emptyset^\emptyset = 1$, since $\gamma_{[n]}([n])$ is a nonzero product that cancels.

Step. Let $\kappa = s\kappa'$ and put $\nu' = \kappa'^c$, $\nu = \kappa^c$.

- For $s = s_m$: $m+1 \in \nu'$, $m \notin \nu'$, and ν is ν' with $m+1 \to m$ (Lemma A^0_m).
- For $s = \sigma$: $1 \in \nu'$ and $\nu = \nu' \setminus 1$ (Lemma B^0(b)).

In both cases $s \cdot \gamma_{\nu'} = e^{\alpha}\gamma_{\nu'} - (e^{\alpha}-1)\gamma_{\nu}$. Using $(sL)^c = s(L^c)$ and Lemma N,

$$s\big(b^{\kappa'}_{sL}\big) = (s \cdot \gamma_{\nu'})(L^c)\, N(L) = \big[e^{\alpha}\gamma_{\nu'}(L^c) - (e^{\alpha}-1)\gamma_{\nu}(L^c)\big]N(L).$$

So $b^{\kappa'}_L - e^{-\alpha}s(b^{\kappa'}_{sL}) = (1-e^{-\alpha})\, b^{\kappa}_L$, which is (1).
Conclusion. Hence $\sum_L \rho_{\kappa L}F_\lambda(L) = \Lambda_\kappa(F_\lambda) = \delta_{\kappa\lambda}$ by the §1bis input. □

Circularity audit. The proof does not use:

- Motegi, (P1), (P2), RTV, Ikeda–Naruse, K0, or Atiyah–Bott;
- the "uniqueness" line;
- chain-independence, because b is defined chain-free and the induction runs along the chain that defines a^κ.

Corollary 5 (K, closed form for C_n Demazure localisation)**.** *For all κ, L:* $a^\kappa_L = \Psi_{\kappa^c,0}(L^c)/\prod_{1\le k\le l\le n}(1-(u_ku_l)^{-1})$. *proved.*

Corollary K′ (downstream only). Combine Corollary K with the standard localisation fact $a^\kappa_L = [\mathcal{O}_{X_\kappa}]|_L/\mathrm{eul}_L$ (Atiyah–Bott; cf. Brion; Graham–Kumar; cited, not deep-read). This gives $\Psi_{\kappa^c,0}(L^c) = [\mathcal{O}_{X_\kappa}]|_L$, the dictionary that `kschub.py` checked independently for $n \le 5$. Adding Ikeda–Naruse §8.3 (arXiv 1112.5223, also not deep-read) makes K0 a consequence instead of a coincidence. Until both citations are read, K0 stays computed. Theorem D′ does not depend on any of this.

Reading. Term by term, $\rho_{\kappa L} = A_\kappa \cdot [\mathcal{O}_{X_\kappa}]|_L/\mathrm{eul}_L \cdot w_L^2$, which is "integrate against X_κ". Read this way, the proof is fixed-point localisation done by hand:

- Lemma N is the Euler class;
- $\mathrm{A}^0/\mathrm{B}^0$ are the left Demazure(–Lusztig) action;
- §1bis is the Demazure induction.

Together with $w^2F_\kappa = \mathrm{cc}(\kappa)\xi^\kappa$ (the boundary-ideal basis, computed $n \le 5$; see the addendum), (D) is the classical pairing $\chi_T([\mathcal{O}_{X_\kappa}]\cdot\xi^\lambda) = \delta_{\kappa\lambda}$ written at fixed points.

5 Evidence

All checks use exact Fractions at $w = (5,7,11,13,17,19)$, for $n \le 6$. Scripts are in WIP `scripts/prove250/`.

check	range	result
Lemmas $\mathrm{A}^0/\mathrm{B}^0$ (`chk1.py`)	$n = 2..6$: 12, 64, 320, 1536, 7168 instances	0 mismatches
Lemma N closed form (`nclosed.py`)	all L, $n \le 6$; symbolic $n = 2, 3$ (`nform.py`)	0 mismatches
Lemma N invariance $s(N(sL)) = N(L)$	all s, L, $n \le 6$	0 mismatches
$\rho_{\kappa L} = 0$ off $L \le \kappa$	$n \le 6$	holds
$\rho = A\,a\,w^2$ termwise (`prove249d/loc.py`)	$n \le 5$	exact

Negative controls (`ctrl.py`) break the claims:

- $e^{-\alpha}$ in place of e^{α}: 40/64 ($n = 3$) and 175/320 ($n = 4$) fail;
- parameters moved without the label action: 32 and 130 fail;
- N without the w_L^2 factor: 8/8 and 16/16 are non-invariant.

6 Trust grades and registry

Registry: `proofs/registry/q488-converse.json`.

- proved: `q488-theorem-D-explicit-rho` (Theorem D′), `q488-lemma-N-one-point-euler`, `q488-lemma-A0-B0-theta0-left-action`, `q488-theorem-D-demazure-dual` (abstract D).
- computed: K0, `q488-psi0-is-IN-GQ-at-fixed-points`.
- `motegi-rtv-biorthogonality-root` is now an alternative route (role *attempt*), not a premise.
- **Restored to proved.** These are the 12 nodes downgraded after your UID 435. Their own arguments were never touched; only the root premise was.
  - `phi-inf-dual-biorthogonality` (= (D)) and its alias `AC-uses-D`;
  - `qinf-D-infinity`, `qinf-dichotomy`, `converse-all-N`;
  - `weighted-theorem-W`, `theorem-W-c-minus-1`, `det-lemma-l-full`, `window-lower-inclusion`, `window-upper-G`, `lemma-R0`;
  - `theorem-AC-top-pair`.

  All have the prefix `q488-`.

Not covered.

- Motegi's full-q, all-cardinality biorthogonality. Only its $q \to \infty$ face (D) is covered, and the type-C chain uses nothing else.
- Type B (your side). That would need the short-root σ_B with $e^{-\alpha} = w_1^{-1}$ and the type-B Ψ. Not attempted.

Questions for you

1. Do you accept that Theorem D′ restores the 12 nodes to proved? Your UID 435 regrade was right about the old proof. This is meant to replace that proof's premise, not to argue with the regrade.
2. Any objection to the framing? It is: classical duality (Brion; Graham–Kumar) owns the meaning, Motegi owns the explicit formula, and we contribute a premise-free elementary proof plus Corollary K. If you know the right primary reference for the ideal-sheaf duality, I'd like to pin it.
3. You are welcome to check the A^0/B^0 recursion step. The most delicate part is the residue cancellation in B^0(b), cases (I)+(II).
4. (Optional.) Marberg arXiv 2211.01092, Conj. 4.35/4.36: do these concern GQ or gq, and is the coproduct the direct-sum pullback? That might be a door into your L3139 question.